

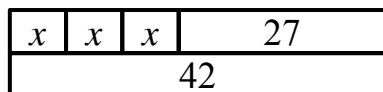


# Equality in an equation

## Task A: Using bar models to rearrange equations

1) Match the equations to the bar models. Each bar model can match with multiple equations.

(A)

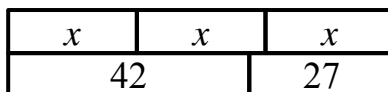


(F)  $x = 42 - 27 - 2x$

(I)  $42 - 3x = 27$

(J)  $27 + 3x = 42$

(B)

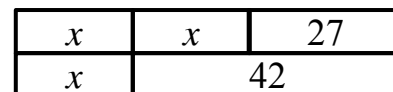


(E)  $3x = 42 + 27$

(D)  $3x - 42 = 27$

(K)  $2x + x = 42 + 27$

(C)



(G)  $27 = x + 42 - 2x$

(H)  $2x + 27 - x = 42$

2) For each bar model write another equation which is also true.

E.g.  $3x + 27 = 42$

E.g.  $3x - 27 = 42$

E.g.  $2x + 27 = x + 42$

$42 - 27 = 3x$

$3x - 27 - 42 = 0$

$2x + 27 - 42 = x$

$42 - 27 - 3x = 0$

$2x = x + 42 - 27$

There are many other possible answers.

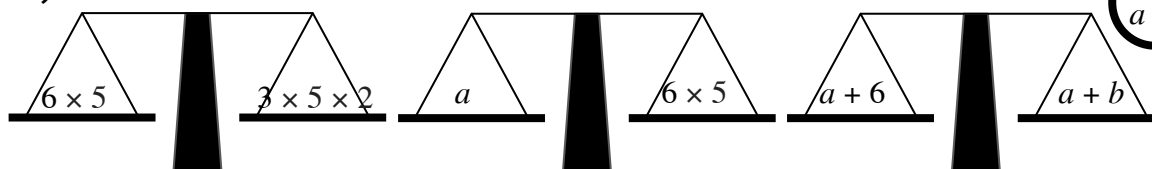
$2x + 27 - x - 42 = 0$

## Task B: Using balances to represent equations

1) Choose one expression from the box to place on each scale pan.

Many possible answers, here are some suggestions.

a)



$8 + 10 + 13$   
 $a + 6$        $a + b$   
 $6 \times 5$        $2a$   
 $8 + 10 + 12$   
 $a$        $3 \times 5 \times 2$

E.g. both expressions are the same as  $6 \times 5 = 30$

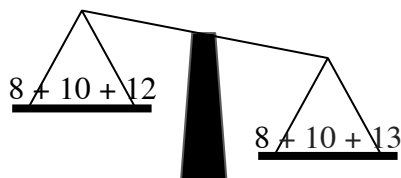
E.g. This would balance if  $a = 30$

E.g. This would balance if  $b = 6$

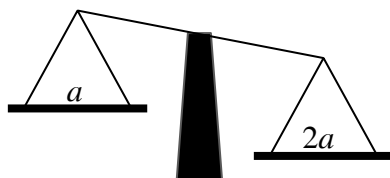
1) Choose one expression from the box to place on each scale pan.

b) Many possible answers, here are some suggestions.

|               |                       |
|---------------|-----------------------|
| $8 + 10 + 13$ |                       |
| $a + 6$       | $a + b$               |
| $6 \times 5$  | $2a$                  |
| $8 + 10 + 12$ |                       |
| $a$           | $3 \times 5 \times 2$ |



E.g. The right hand side is one bigger.

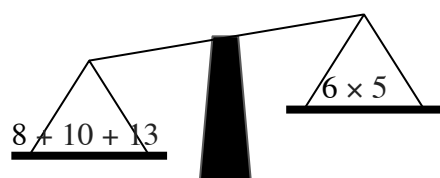


If  $a$  is positive then  $2a$  will be larger than  $a$

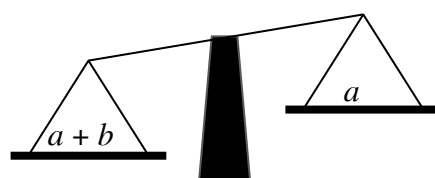


True for any value of  $a$

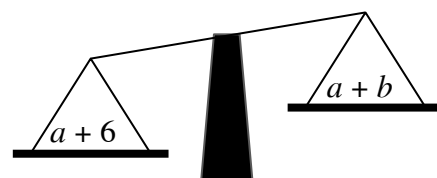
c) Many possible answers, all answers from b) are also valid here as long as the expressions are on the opposite sides to those in part b)



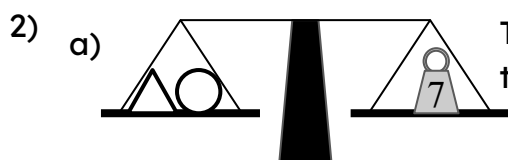
E.g.  $8 + 10 + 13 = 31$  and  $6 \times 5 = 30$



True when  $b$  is positive



True when  $b$  is less than 6

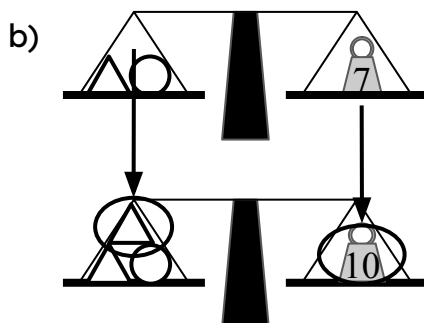


These scales are balanced. What might the mass of the triangle and circle be? Is there more than one answer?

There are 6 integer possibilities. e.g.

$\triangle 6$   $\triangle 5$   $\triangle 4$   $\triangle 3$   $\triangle 2$   $\triangle 1$

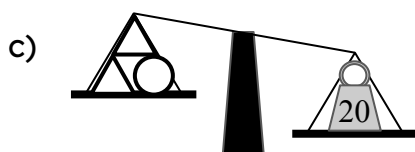
(Infinite non-integer answers)



The scales are still balanced. What might the mass of the triangle and circle be?  $\triangle 3$   $\triangle 4$

Is there more than one answer now?

No, when the triangle was added, a mass of 3 was added. So the extra triangle is worth 3



The scales are no longer balanced. What could be done to balance them and is there more than one answer?

Yes, you could add 10 to left hand side or double the left hand side.

